

PFIM 7.1 Beta version — Test and Optimisation Scripts for covariate and IOV

This folder contains five R scripts for testing and optimising population pharmacokinetic designs using the PFIM package. All scripts share the same underlying model: a one-compartment PK model with first-order absorption, log-normally distributed parameters, and a constant additive residual error.

Common model specification

Element	Value
PK equation	$\text{dose}/V * ka / (ka - Cl/V) * (\exp(-Cl/V*t) - \exp(-ka*t))$
ka	LogNormal(mu = 1, omega = 0.3)
V	LogNormal(mu = 3.5, omega = 0.3)
Cl	LogNormal(mu = 2 or 0.5, omega = 0.3)
IOV (when present)	gamma = sqrt(0.0225) on all three parameters
Residual error	Constant, sigmaInter = 0.1
Fixed covariate	Sex on V: beta = log(1.2), ratio V_F/V_M = 1.20, 50/50 M/F
Occasion covariate	Treatment on Cl: AB/BA cross-over, beta = log(1.1), ratio Cl_B/Cl_A = 1.10

Script descriptions

`script_tests_cov_and_iov_and_tests_tost_equi_noequiv.R`

Runs six successive FIM evaluation cases that progressively increase model complexity. Each case calls `run()` on an `Evaluation` object, displays the result with `show()`, and saves the FIM, D-criterion, and standard errors to a text file via `saveResult()`. Cases that include covariates additionally call `covariateTest()` and save the three resulting statistical tables (significance, non-relevance, relevance) via `saveResultCovariateTest()`.

The six cases are:

Case	Covariates	IOV
1	None	No
2	None	Yes (2 occasions)
3	Fixed: Sex on V	No
4	Fixed: Sex on V	Yes (2 occasions)
5	Fixed: Sex on V + occasion: Treatment on Cl	Yes (2 occasions)
6	Occasion: Treatment on Cl only	Yes (2 occasions)

Each case is isolated with `rm()` + `devtools::load_all()` before execution to ensure a clean environment.

`optimisation_algo_mult_withCov_mixed.R`

Optimises sampling times using the **Multiplicative algorithm** for a model with both a fixed covariate (Sex on V) and an occasion covariate (Treatment on Cl, AB/BA cross-over, 2 occasions). The design has 1 arm of

40 subjects with 5 candidate sampling times (0.5, 2, 4, 6, 8 h), of which 3 are optimisable. The algorithm maximises the D-criterion of the population FIM.

Optimiser parameters: $\lambda = 0.99$, 1000 iterations, weight threshold = 0.01, $\delta = 1e-4$.

`optimisation_algo_simplex_withCov_mixed.R`

Optimises sampling times using the **Simplex algorithm** for the same mixed covariate structure as above (Sex on V + Treatment on Cl, AB/BA cross-over, 2 occasions). The design has 1 arm of 100 subjects with 10 candidate sampling times distributed across two windows: [0, 24] h (6 samples) and [35, 130] h (4 samples), with minimum inter-sample gaps of 1 h and 2 h respectively.

Optimiser parameters: 20% initial simplex perturbation, 200 iterations, tolerance = $1e-6$.

`optimisation_algo_simplex_withCov_withIOV.R`

Optimises sampling times using the **Simplex algorithm** for a model with a fixed covariate only (Sex on V) and IOV on all three parameters (2 occasions). Same design and constraint structure as `optimisation_algo_simplex_withCov_mixed.R` (100 subjects, two sampling windows), but without the occasion covariate Treatment.

`optimisation_algo_simplex_withCovOccasion.R`

Optimises sampling times using the **Simplex algorithm** for a model with an occasion covariate only (Treatment on Cl, AB/BA cross-over, 2 occasions) and no fixed covariate. Same design and constraint structure as the other Simplex scripts (100 subjects, two sampling windows).

Output files

All evaluation results are written to the directory defined by `file.path(folder_PFIM, folder_results)` at the top of the evaluation script. Each case produces one or two plain-text files:

- `casN_<label>.txt` — FIM matrix, D-criterion, and SE for the evaluation.
- `casN_<label>_resultsClinicalRelevance.txt` — significance, non-relevance, and relevance tables from `covariateTest()` (cases 3–6 only).

Optimisation scripts display results in the console via `show()` and do not write output files.

Dependencies

- R package **PFIM** (loaded via `library(PFIM)` or `devtools::load_all()`).
- No other external packages are required beyond those imported by PFIM.